METRIC

FF-B-171/6

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, ANSI/AFBMA DIMENSION SERIES 04.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

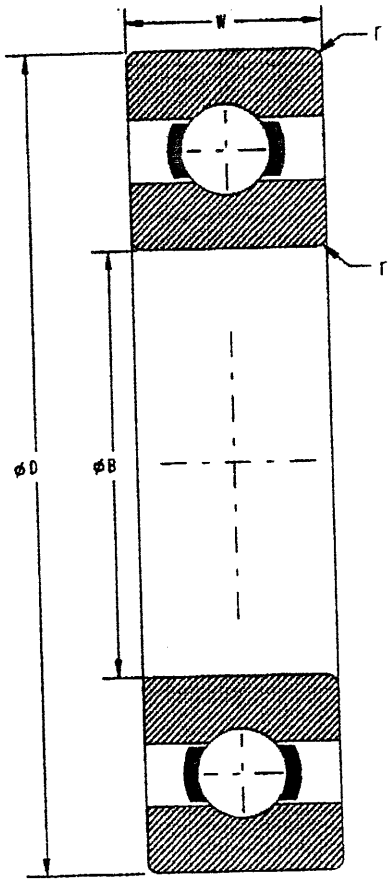


FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)

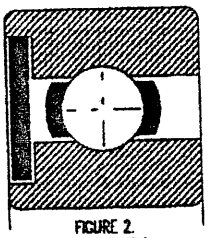


FIGURE 2.
SINGLE SHIELD
(CODE B)

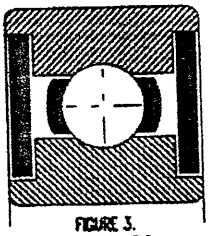


FIGURE 3.
DOUBLE SHIELD
(CODE C)

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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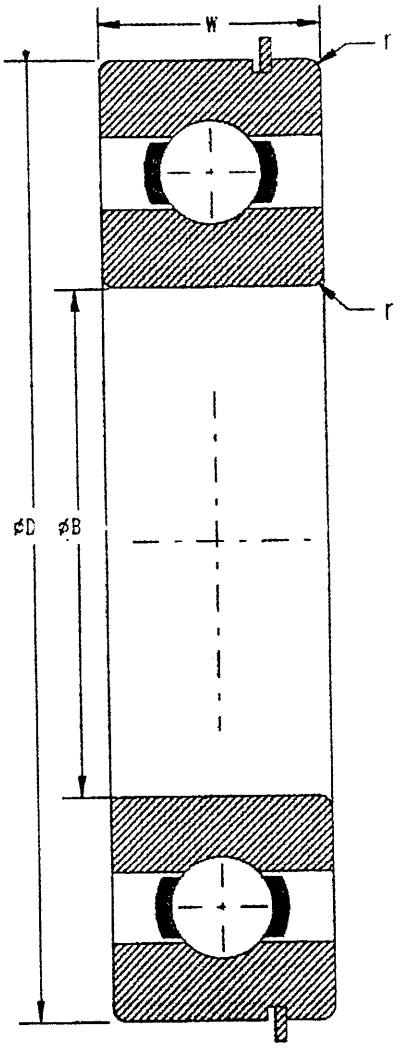


FIGURE 4.
OPEN SNAP RING
(CODE G)

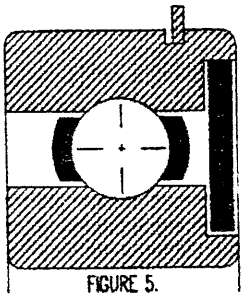


FIGURE 5.
SINGLE SHIELD-
ATTACHED ON SNAP
RING SIDE
(CODE H)

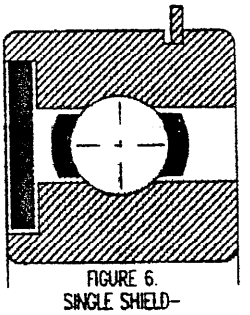


FIGURE 6.
SINGLE SHIELD-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE J)

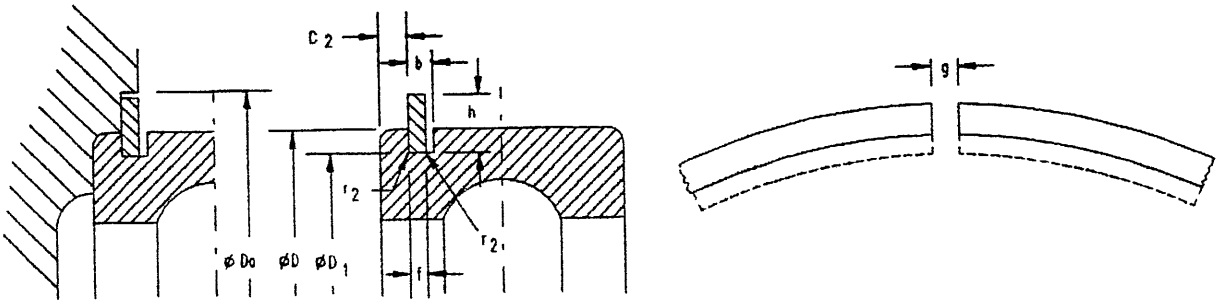


FIGURE 7.
SNAP RING AND SNAP RING GROOVE DIMENSIONS

ØD =BASIC BEARING OUTSIDE DIAMETER
ØD 1 =SNAP RING GROOVE DIAMETER
C 2 =SNAP RING GROOVE LOCATION
b =SNAP RING GROOVE WIDTH
h =SNAP RING SECTION HEIGHT
f =BASIC RING THICKNESS
g =GAP OF MOUNTED SNAP RING
r 2 =FILLET RADII AT SNAP RING GROOVE BOTTOM
Do =DIAMETER OF END COVER COUNTERBORE

BEARING CONFIGURATIONS - (Continued)

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2* Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields and snap rings. Shields shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields and snap rings required shall be specified in the part number by the codes in table III.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

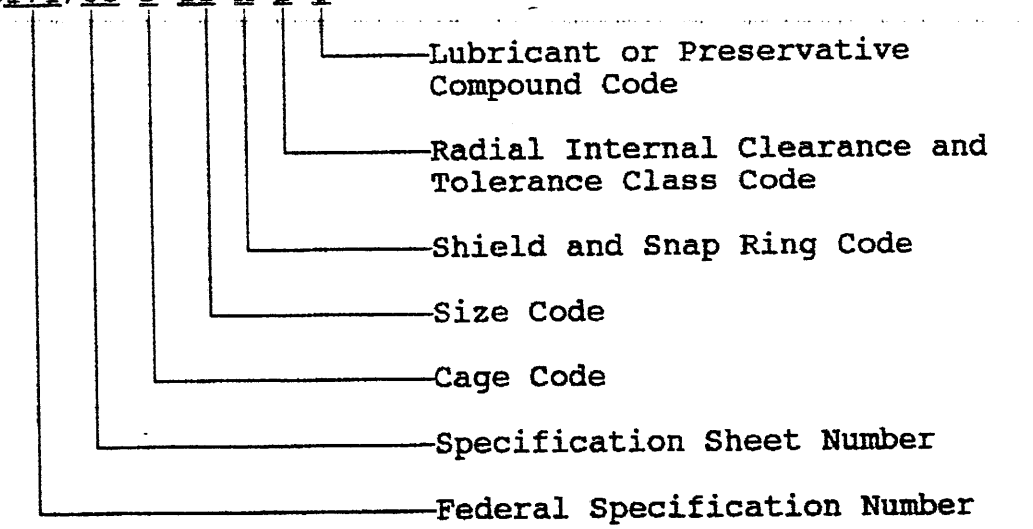
5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Lubricant Preservative Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the “A” or “B” suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/06 J 11 A 2 C



FFB171/06J11A2C indicates - Bearing, cage-pressed steel or molded plastic; bore 55mm: outside diameter.. 140mm; width 33mm; open (no shields or seals); radial internal clearance symbol 0 (normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-1970

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic Load Rating (Pounds)
01	12	42	13	1.0	16.3	2010
02	15	52	15	1.1	20.3	2730
03	17	62	17	1.1	22.4	3830
04	20	72	19	1.1	25.7	5285
05	25	80	21	1.5	32	6165
06	30	90	23	1.5	37.3	7460
07	35	100	25	1.5	42.4	9440
08	40	110	27	2.0	49.5	10990
09	45	120	29	2.0	54.6	13324
10	50	130	31	2.1	60.5	15110
11	55	140	33	2.1	66	17355
12	60	150	35	2.1	71.4	18890
13	65	160	37	2.1	76.5	20455
14	70	180	42	3.0	83.8	24830
15	75	190	45	3.0	88.9	26530
16	80	200	48	3.0	94	28260
17	85	210	52	4.0	102.6	30015
18	90	225	54	4.0	108	31915
19	95	240	55	4.0	113	32000
20	100	250	58	4.0	118.1	35000

1/ Size codes in Table I are industry bore codes.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)

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TABLE III. Codes for Shields and Snap Rings.

Code	Shield and snap ring configuration
A	open (no shields or seals) - fig.1
B	single shield - fig.2
C	double shield - fig.3
G	open with snap ring - fig.4
H	single shield (attached on snap ring side) - fig.5
J	single shield (attached on side opposite snap ring)-fig.6

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

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